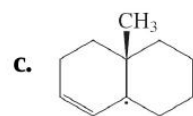
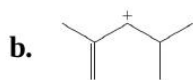
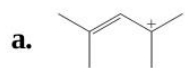


32. Draw all resonance forms and a representation of the appropriate resonance hybrid for each of the following species.

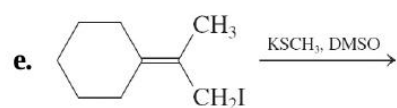
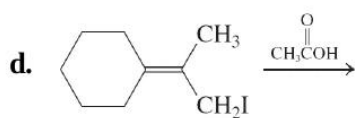
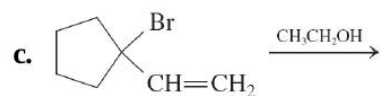
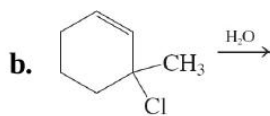
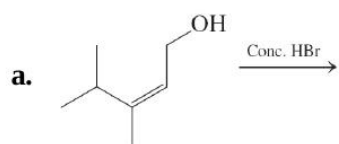


33. For each species in [Problem 32](#), indicate the resonance form that is the major contributor to the resonance hybrid. Explain your choices.

34. Illustrate by means of appropriate structures (including all relevant resonance forms) the initial species formed by **(a)** breaking the weakest C—H bond in 1-butene; **(b)** treating 4-methylcyclohexene with a powerful base (e.g., butyllithium-TMEDA); **(c)** heating a solution of 3-chloro-1-methylcyclopentene in aqueous ethanol.

35. Rank primary, secondary, tertiary, and allylic radicals in order of decreasing stability. Do the same for the corresponding carbocations. Do the results indicate something about the relative ability of

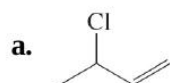
36. Give the major product(s) of each of the following reactions.



37. Formulate detailed mechanisms for the reactions in [Problem 36](#), parts (a), (c), (e), and (f).

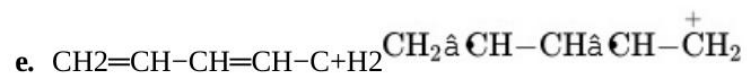
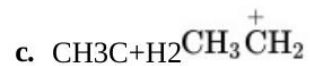
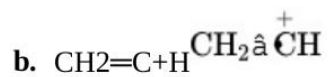
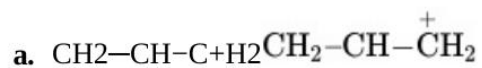
38. Rank primary, secondary, tertiary, and (primary) allylic chlorides in approximate order of **(a)** decreasing $\text{S}_{\text{N}}1$ reactivity; **(b)** decreasing $\text{S}_{\text{N}}2$ reactivity.

39. Rank the following six molecules in approximate order of decreasing $\text{S}_{\text{N}}1$ reactivity and decreasing $\text{S}_{\text{N}}2$ reactivity.



40. How would you expect the $\text{S}_{\text{N}}2$ reactivities of simple saturated primary, secondary, and tertiary chloroalkanes to compare with the $\text{S}_{\text{N}}2$ reactivities of the compounds in [Problem 39](#)? Make the same comparison for $\text{S}_{\text{N}}1$ reactivities.

54. Arrange the following carbocations in order of decreasing stability.
Draw all possible resonance forms for each of them.



60. Give the product(s) of each of the following reactions.

- a. 3-Chloro-1-propene (allyl chloride) + NaOCH₃ ^{NaOCH₃}
- b. *cis*-2-Butene + NBS, peroxide (ROOR)
- c. 3-Bromocyclopentene + LDA
- d. *trans,trans*-2,4-Hexadiene + HCl ^{HCl}
- e. *trans,trans*-2,4-Hexadiene + Br₂, H₂O ^{Br₂, H₂O}
- f. 1,3-Cyclohexadiene + methyl propenoate (methyl acrylate)
- g. 1,2-Dimethylenecyclohexane + methyl propenoate (methyl acrylate)

79. Arrange the following three chlorides in decreasing order of SN_1 reactivity.



A



B



C

- a. $\text{A} > \text{B} > \text{C}$; $\text{A} > \text{B} > \text{C}$;
- b. $\text{B} > \text{C} > \text{A}$; $\text{B} > \text{C} > \text{A}$;
- c. $\text{B} > \text{A} > \text{C}$; $\text{B} > \text{A} > \text{C}$;
- d. $\text{C} > \text{B} > \text{A}$; $\text{C} > \text{B} > \text{A}$.